

MD MAHMODUR RAHMAN

Software Developer | AI & Full-Stack Applications

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PROFILE

Results-driven Computer Science and Engineering undergraduate specializing in AI-integrated web applications, with hands-on experience building RAG pipelines, vector databases, and LLM-powered tools. Proven ability to ship production-quality, full-stack software rapidly, including AI-assisted “vibe coding” workflows, alongside solid contributions to .NET enterprise applications. Currently serving as Public Relations Officer for IEEE Computer Society, IIUC Student Branch Chapter, combining technical depth with strong leadership and communication skills.

PROJECTS

NewsNext — AI-Powered Authentic News Aggregation & Filtering Platform

- Engineered an AI-driven platform that processes real-time news data to surface authentic, relevant stories, leveraging the Gemini API and Grok for NLP-based content verification.
- Implemented LangChain with a vector database and Retrieval-Augmented Generation (RAG) to significantly improve content relevance and factual accuracy.
- Designed a modern, responsive UI optimized for seamless, intuitive user interaction.

Doctor's AI — AI Assistant for Summarizing Medical Reports

- Built a tool enabling doctors to upload patient report PDFs and instantly receive concise, AI-generated summaries, cutting report review time substantially.
- Applied LLM-based document parsing and summarization techniques, prioritizing clinical accuracy and clarity to support faster decision-making.

Bachelor's Point — Full-Featured .NET Web Application

- Architected and developed a complete web application in .NET (C#) using a clean, layered architecture (Controllers, Services, Repositories, Models, Data, Unit of Work).
- Implemented Google OAuth login/registration, SSLCommerz payment integration, an admin approval workflow, and pagination; collaborated with a co-developer across 38+ commits.

B-FIM – Blockchain-Based File Integrity Monitor — Windows Desktop App (C#, .NET 9, WPF)

- Built a real-time file integrity monitor that fingerprints files with SHA-256 and records every event in a tamper-evident, blockchain-style SQLite ledger with chained block hashing.
- Implemented core OS algorithms (Round Robin scheduling, LRU page replacement, SSTF disk scheduling); added ECDSA P-256 signatures, dual chain/file verification, ACL/SID access tracking, and a WPF dashboard.

RESEARCH & LEADERSHIP

- Co-Author of one research paper (awaiting publication); currently conducting an additional ongoing research project.
- Public Relations (PR) Officer, IEEE Computer Society (CS) – IIUC Student Branch Chapter, driving member engagement and event communications.

EDUCATION

B.Sc. in Computer Science and Engineering (CSE)

International Islamic University Chittagong (IIUC)

Higher Secondary School Certificate (HSC)

Hazi Mohammad Mohsin College

Secondary School Certificate (SSC)

Collegiate School

SKILLS

Languages: C#, Python, Java, JavaScript, HTML5, SQL.

AI & Data: Gemini API, Grok, LangChain, Vector Databases, RAG (Retrieval-Augmented Generation), AI Integration

Frameworks & Tools: .NET Core, React, Version Control (GitHub), AI-Assisted Development

Soft Skills: Public Relations, Team Leadership, Event Management, Strategic Communication, Collaboration

Language: IELTS – Band 6.5 (2024)